

Interpolation-based parametric reduced-order models with dynamic mode decomposition

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ABSTRACT

Dynamic mode decomposition (DMD), a purely data-driven approach, is extensively applied to the construction of reduced-order models (ROMs) and holds significant potential for the development of parametric reduced-order models (PROMs). To construct accurate and efficient data-driven DMD PROMs, this work integrates various interpolation methods with the DMD framework to interpolate both DMD modes and eigenvalues for target (unseen) parameters. Specifically, three interpolation approaches are proposed for complex-valued, non-orthonormal DMD modes: (1) subspace angle interpolation (SAI), (2) Grassmann manifold interpolation (GMI), and (3) mode-realigned pointwise interpolation (MRPWI). These interpolation methods—the SAI (based on the concept of principal angles), the GMI (formulated on the complex Grassmann manifold via the compact Stiefel manifold representation), and the MRPWI (based on Kasner's pseudo-angle)—provide accurate approaches for constructing PROMs. Meanwhile, the eigenvalues are determined exclusively via Lagrange interpolation. Additionally, the mode ranking criterion—combining the existing mode ranking criterion with imaginary part ranking—is innovatively formulated and incorporated into these interpolation methods, where it serves as a critical component for accurate construction of PROMs. Flow over a cylinder serves as the numerical case for demonstrating and evaluating the developed PROMs, with particular focus on the evolution of flow fields and aerodynamic coefficients. With direct numerical simulations (DNS) and ROMs as benchmarks, the PROMs constructed with the proposed interpolation methods demonstrate significantly better performance compared to existing strategies. This shows that the proposed PROMs can effectively serve as superior substitutes for ROMs when flow data at target parameters are unavailable. Specifically, the PROMs constructed via MRPWI are particularly notable owing to the theoretical simplicity, computational efficiency, and scalability inherent in MRPWI.

1. Introduction

Reduced-order models (ROMs) are widely utilized in science and engineering as efficient and accurate tools for approximating full-order models (FOMs) within low-rank subspaces [1]. ROMs preserve the dominant features of high-dimensional, complex dynamical systems while avoiding the high computational costs required by FOMs [2]. This enables significant improvements in computational speed while maintaining the desired level of accuracy. Indeed, various dimensionality reduction techniques have

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been developed for constructing ROMs [3], such as proper orthogonal decomposition (POD) [4,5], dynamic mode decomposition (DMD) [6], locally linear embedding [7], and isometric feature mapping [8]. A comprehensive review and comparison of these techniques are provided in Taira et al. [9], Rowley and Dawson [10]. Among these methods, DMD has emerged as a particularly powerful tool because of its equation-free nature and capability to extract spatiotemporal coherent structures from complex system data [11].

DMD is a purely data-driven algorithm designed to extract the best-fit linear dynamical system evolving in time from high-dimensional data, without requiring prior knowledge of governing equations [11,12]. It integrates POD for spatial dimensionality reduction with complex-valued temporal frequency identification and is widely employed in fields such as fluid mechanics, video processing, and others [2,13]. Originally developed by Schmid [6] to analyze high-dimensional flow data, DMD enables the decomposition of complex flows into spatiotemporal coherent structures. Subsequently, Rowley et al. [14] clarified an important connection between DMD and Koopman theory, demonstrating that DMD identifies a finite-dimensional approximation of the Koopman operator. To date, numerous DMD variants have been developed for different applications [3], such as optimized DMD [15], sparsity-promoting DMD [16], higher-order DMD [17], and physics-informed DMD [18]. These extensions aim to enhance the accuracy, robustness, and efficiency of DMD, while leveraging advancements from other fields to further refine its core methodology [19].

Although ROMs are highly accurate and efficient in representing dynamical systems, they struggle with parameter changes in parametric dynamical systems. These systems are more prevalent in practical applications and often incur prohibitive computational costs as parameters vary. To address this limitation, parametric reduced-order models (PROMs) have been developed within the DMD framework. PROMs extend traditional ROMs by incorporating parameter dependencies within the dynamical system and can be constructed via interpolation-based or stacking-based methods [20]. Huhn et al. [21] applied Lagrange interpolation to the entries of either the low-dimensional linear operator or its eigenvalues and eigenvectors to construct DMD PROMs, which are more efficient and better suited for multi-parametric problems compared to stacked parametric DMD [22]. However, as illustrated in Benner et al. [20], Amsallem [23], straightforward entry-by-entry interpolation cannot preserve the desired intrinsic properties of the variables involved in this interpolation process. Specifically, the interpolated quantities may not lie on the specified manifolds, often resulting in inaccurate or meaningless results [20]. To develop DMD PROMs, Hess et al. [24] evaluated the low-dimensional linear operator and POD modes for target parameters, employing the concepts of the general linear group and Grassmann manifold for interpolation, respectively. These models were then applied to investigate thermal convection flows exhibiting multiple Hopf bifurcations. Nevertheless, the requirement of inconsistent interpolation methods for various quantities introduces theoretical complexity and computational expense, potentially increasing numerical errors during the interpolation process. In addition, Li et al. [25] utilized radial basis functions (RBFs) within the DMD framework for parameterized problems. Furthermore, Gao et al. [26] and Song et al. [27] incorporated k -nearest-neighbors (KNN) regression and weighted interpolated nearest-neighbors (wiNN) algorithms, respectively, into the DMD framework. These approaches enabled the development of PROMs for nonlinear parameterized partial differential equations (PDEs), with both demonstrating excellent performance. These studies demonstrate the effectiveness of embedding machine learning techniques (RBFs, KNN, wiNN) into the DMD framework for constructing PROMs. However, the inherent trade-offs between computational tractability and model generalizability necessitate rigorous analysis, particularly in high-dimensional multiscale systems or strong nonlinear regimes.

Inspired by prior research, we aim to develop accurate and efficient DMD PROMs by integrating various interpolation methods with DMD modes and eigenvalues. The principal angles between subspaces describe the rotation angles of corresponding principal vectors, providing a basis for the subspace angle interpolation (SAI) method [28]. This approach facilitates the interpolation of subspace angles to obtain the variables corresponding to the target parameter located between the two subspaces [29,30]. Whereas the prior SAI method has been limited to real-valued subspaces, this work generalizes it to complex-valued subspaces. Furthermore, Grassmann manifold interpolation (GMI) involves projecting the data points onto the tangent space of a reference point on the manifold, performing interpolation within this tangent space, and then mapping the interpolated results back to the manifold. This approach not only preserves the desired intrinsic properties of the quantities throughout the process but also facilitates high-order interpolation, enabling the construction of more accurate and robust PROMs [31]. Although earlier studies have employed GMI for constructing PROMs, they have exclusively considered real-valued interpolation. Here, we rigorously establish the framework and theoretical foundation for GMI on the complex Grassmann manifold via the compact Stiefel manifold representation. Additionally, a highly efficient interpolation algorithm derived from Kasner's pseudo-angle-mode-realigned pointwise interpolation (MRPWI)-is proposed, with rigorous mathematical foundations. This approach enables high-order interpolation without relying on orthonormality constraints, in contrast to the requirements of SAI and GMI. The SAI and GMI, both combined with the modified Gram-Schmidt (MGS) algorithm, as well as the MRPWI, are applied to the complex-valued, non-orthonormal DMD modes to construct PROMs. The corresponding eigenvalues, without extra intrinsic properties, are interpolated with straightforward approaches. Moreover, the DMD modes and eigenvalues are sorted beforehand according to an improved mode ranking criterion. This criterion combines the existing mode ranking criterion with sorting by imaginary parts, and it is incorporated into the developed interpolation methods. Finally, the performance of the proposed PROMs, along with existing methods, is evaluated and compared using a two-dimensional flow over a cylinder.

The remainder of this paper is organized as follows: Section 2 introduces the framework of DMD. Section 3 details the interpolation methods and their applications. In Section 4, a test case in fluid dynamics is presented, including analyses of flow fields and aerodynamic coefficients. Section 5 summarizes the present work and provides concluding remarks. In addition, Appendix A and Appendix B establish the mathematical framework for complex-valued GMI and MRPWI, respectively.

2. Dynamic mode decomposition

A sequence of snapshots $\{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{N_s}\} \subset \mathbb{C}^{N_n}$ is collected from a nonlinear dynamical system as it evolves over time. These snapshots are high-dimensional column vectors, with each snapshot $\mathbf{x}_i$ representing the state vector of the dynamical system at time t_i . The data matrices X and X' are then defined as follows:

$$X = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{N_s-1}] \in \mathbb{C}^{N_n \times N_s}, \quad (1a)$$

$$X' = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_s}] \in \mathbb{C}^{N_n \times N_s}. \quad (1b)$$

In most applications, the spatial dimensionality is significantly larger than the number of snapshots, i.e., $N_n \gg N_s$. For example, a snapshot $\mathbf{x}_i$ of a two-dimensional incompressible flow can be expressed as:

$$\mathbf{x}_j = [u(t_j), v(t_j), p(t_j)]^T \in \mathbb{R}^{N_n}, \quad (2)$$

where $u(t_j)$, $v(t_j)$ and $p(t_j)$ represent the velocity components and pressure of the flow fields, respectively, reshaped into a column vector.

The nonlinear dynamical system can be approximated linearly in terms of data matrices:

$$X' \approx AX, \quad (3)$$

with the best-fit linear operator A that serves as a finite-dimensional approximation of the Koopman operator [2,11] and generates a linear dynamical system advancing the data snapshots forward in time. The operator A can be obtained by minimizing the Frobenius norm:

$$A = \arg \min_A \|X' - AX\|_F = X'X^\dagger, \quad (4)$$

where $\|\cdot\|_F$ and $\dagger$ are the Frobenius norm and the Moore-Penrose pseudoinverse, respectively.

DMD focuses on extracting spatio-temporal modes, i.e., the eigenvectors and eigenvalues of the linear operator A , as well as the amplitudes that denote the contribution of the modes to the initial state of the dynamical system [32]. Since the high-dimensional linear operator A is intractable to form and analyze explicitly, the DMD algorithm leverages a low-dimensional linear operator $\tilde{A}$ to approximate A . A detailed description of the DMD algorithm is provided below:

1. Conduct the singular value decomposition (SVD) of X and truncate it to obtain a rank- N_r approximation ($N_r \leq \min\{N_s, N_n\}$):

$$X \approx U\Sigma V^*, \quad (5)$$

where $*$ denotes the conjugate transpose, $U \in \mathbb{C}^{N_n \times N_r}$, $\Sigma \in \mathbb{C}^{N_r \times N_r}$, and $V \in \mathbb{C}^{N_s \times N_r}$. The columns of the matrix U are known as the left singular vectors, or POD modes, and both the left and right singular vectors are orthonormal, i.e., $U^*U = I$ and $V^*V = I$. The singular values contained in the diagonal matrix Σ are non-negative and decrease sharply toward zero.

2. The linear operator A can be calculated from Eqs. (3) and (5):

$$A = X'(U\Sigma V^*)^\dagger = X'V\Sigma^{-1}U^*. \quad (6)$$

Then, the low-dimensional linear operator $\tilde{A}$ can be obtained by projecting the matrix A onto the truncated POD modes:

$$\tilde{A} = U^*AU = U^*X'V\Sigma^{-1} \in \mathbb{C}^{N_r \times N_r}. \quad (7)$$

3. Perform the eigendecomposition of $\tilde{A}$:

$$\tilde{A}W = W\Lambda, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_{N_r}), \quad (8)$$

where the columns of the matrix $W \in \mathbb{C}^{N_r \times N_r}$ are the eigenvectors, and the entries of the diagonal matrix Λ are the corresponding eigenvalues, also referred to as discrete-time eigenvalues. Following this, the continuous-time eigenvalues are computed as:

$$s_j = \ln \lambda_j / \Delta t, \quad (9)$$

where the real part of s_j denotes exponential growth or decay, and the imaginary part represents the temporal frequency [33]. Notably, the non-zero eigenvalues of the matrices A and $\tilde{A}$ are identical [34].

4. Compute the projected DMD modes:

$$\Phi = UW \in \mathbb{C}^{N_n \times N_r}, \quad (10)$$

where the columns ϕ_j of the matrix Φ represent the DMD modes. For real-valued flow data, each DMD mode associated with a specific eigenvalue $\lambda = a + ib$ exhibits oscillatory behavior while either growing or decaying. Consequently, the eigenvalues are either real or appear as complex conjugate pairs, expressed as $\lambda = a \pm ib$ ($b \neq 0$) represents the same spatial structure with a phase difference, commonly referred to as conjugate modes [6,35].

5. Finally, the evolution of the dynamical system is approximated by:

$$\mathbf{x}(t) = \Phi \exp(S t) \Phi^\dagger \mathbf{x}_0. \quad (11)$$

The operation $\exp(S t)$ is taken elementwise on the diagonal elements of the matrix S , where $S = \text{diag}(s_1, \dots, s_{N_r})$.

3. Interpolation methods and their applications

3.1. Applications of interpolation methods

Following established methodologies, the system evolution equation (Eq. (11)) at the target parameter $\bar{\gamma}$ has been constructed by interpolating either (i) the POD modes U and low-dimensional linear operator $\bar{A}$, or (ii) the POD modes U , eigenvectors W , and eigenvalues Λ [21,24,36]. Since $\Phi = UW$, we notice that directly interpolating DMD modes Φ and eigenvalues S is mathematically more straightforward and has the potential to reduce the cumulative interpolation error. Therefore, this work develops novel interpolation schemes-including subspace angle interpolation (SAI), Grassmann manifold interpolation (GMI), and mode-realigned pointwise interpolation (MRPWI)-specifically designed to handle complex-valued, non-orthonormal DMD modes. Additionally, the continuous-time eigenvalues s_j (instead of the discrete-time eigenvalues λ_j) are interpolated in the complex plane. This approach ensures proper interpolation of both the modulus and phase of complex-valued eigenvalues, particularly for 2-point interpolation methods. The procedures for constructing DMD PROMs are outlined in Algorithm 1.

Algorithm 1 Construction of DMD PROMs.

Input: Flow data $\{X_j \in \mathbb{C}^{N_n \times N_s}\}_{j=1}^{N_c}$ and $\{X'_j \in \mathbb{C}^{N_n \times N_s}\}_{j=1}^{N_c}$, associated with system parameters $\{\gamma_j\}_{j=1}^{N_c}$.

Output: Predicted state $x(t)$ at the target parameter $\bar{\gamma}$.

- 1: Extract the DMD modes $\{\Phi_j \in \mathbb{C}^{N_n \times N_r}\}_{j=1}^{N_c}$ and eigenvalues $\{S_j \in \mathbb{C}^{N_r \times N_r}\}_{j=1}^{N_c}$ via standard DMD, corresponding to available cases with parameters $\{\gamma_j\}_{j=1}^{N_c}$. [Section 2](#)
 - 2: Sort each Φ_j and S_j according to the mode ranking criterion. [Section 3.2](#)
 - 3: Identify N_p neighboring cases from the initial set of N_c cases, then select the reference case ($j = 0$) from these N_p cases if needed. [Section 3.3](#)
 - 4: Interpolate the DMD modes $\bar{\Phi}$ at $\bar{\gamma}$ with methods such as SAI, GMI, or MRPWI from the modes $\{\Phi_j\}_{j=1}^{N_p}$. [Sections 3.4, 3.5, 3.6](#)
 - 5: Determine the eigenvalues $\bar{S}$ at $\bar{\gamma}$ via Lagrange interpolation from the eigenvalues $\{S_j\}_{j=1}^{N_p}$.
 - 6: Predict the evolution of the dynamical system as: $x(t) = \bar{\Phi} \exp(\bar{S}t) \bar{\Phi}^\dagger x_0$.
-

Specifically, the improved mode ranking ([Section 3.2](#)) serves as an important preconditioning method to avoid mismatches between DMD modes and eigenvalues. Additionally, both SAI and GMI necessitate the orthonormality of the quantities involved in the interpolation process [28,37]. Therefore, the modified Gram-Schmidt (MGS) algorithm is employed to orthonormalize the DMD modes. Furthermore, Lagrange interpolation is not only employed for interpolating $\bar{S}$ but is also integrated into SAI, GMI, and MRPWI.

The accuracy of PROMs depends not only on the selection of N_p interpolating cases but also on the choice of a reference case from among them. The subspace distance described in [Section 3.3](#) is likely to sort out neighboring cases that provide better accuracy, but it requires considerable computational cost. Therefore, in practice, a simplified rule can be used to reduce the reliance on subspace distance calculations. Specifically, we first select N'_p cases (where N'_p is even) with parameter values γ closest to the target $\bar{\gamma}$, thus avoiding subspace distance calculations. Here, $N'_p = N_p$ if N_p is even, and $N'_p = N_p - 1$ if N_p is odd. Then, if needed, the case with the γ value closest to $\bar{\gamma}$ is selected to also serve as the reference case ($j = 0$). Additionally, when N_p is odd, the final case j_{N_p} is selected from the remaining candidates as the one with the minimal subspace distance to the reference case.

3.2. Mode ranking

To the best of our knowledge, the development of mode ranking and selection criteria focuses on compactly and optimally extracting dominant modes. These dominant modes are typically identified based on the norm of the modes [14,38], or the initial amplitudes of normalized modes [16,34,35,39]. Furthermore, Sayadi et al. [22], Tissot et al. [40], Kou and Zhang [41], Xu et al. [42] have incorporated the temporal evolution of modes into their ranking criteria. These criteria are also effective in constructing PROMs.

Here, the ranking method developed by Xu et al. [42] is employed, while the sampling time interval Δt is omitted as it remains constant, similar to the approach in Weiner and Semaan [43]. Furthermore, we propose the following rule: for each pair of complex conjugate eigenvalues and the corresponding DMD modes, the ordering is determined by the sign of the imaginary parts of the eigenvalues (from positive to negative), ensuring optimal interpolation performance.

Specifically, a discretized ranking criterion is formulated by systematically considering the temporal evolution of modes that exhibit multiple growth/decay rate behaviors. The criterion is given by:

$$I_j \approx \sum_{k=1}^{N_s} \|\phi_j\|_1 |b_{jk}|, \quad (12)$$

where $\|\phi_j\|_1$ denotes the 1-norm of the j th DMD mode. Additionally, $|b_{jk}|$ represents the absolute value of the mode time coefficient contained in the matrix $B = W^{-1} \Sigma V^*$ corresponding to the j th mode. Here, W , Σ , and V are the eigenvectors, singular values, and right singular vectors, respectively, as described in [Section 2](#).

Mode ranking is applied in all proposed interpolation methods: the DMD modes and eigenvalues are first ranked and then used to generate the corresponding quantities for the target parameter. The benefits of mode ranking as a preconditioning procedure are demonstrated in [Section 4.2](#).

3.3. Distance between subspaces

There are several distance metrics between subspaces, which are detailed in Edelman et al. [44], Giovanis and Shields [45]. Here, the subspaces are spanned by the columns of matrices, and the distance metric is defined as:

$$\text{dist}(\Omega_1, \Omega_2) = \|\cos^{-1} H\|_2, \quad H = \text{diag}(\delta_1, \dots, \delta_{N_r}), \quad (13)$$

where δ_j are the singular values of $\hat{\Omega}_1^* \hat{\Omega}_2$, with $\hat{\Omega}_j$ denoting the corresponding orthonormal matrices obtained by applying the MGS procedure to their column vectors. The algebraic operations involving matrices denote matrix functions (not elementwise operations), with this convention applying to all subsequent expressions unless explicitly stated otherwise. This distance metric is employed in the GMI, PWI, and MRPWI methods for neighboring case identification. Additionally, it corresponds to the geodesic distance—the shortest path between any two points on the Grassmann manifold $\mathcal{G}$.

3.4. Subspace angle interpolation

The subspace angle is a concept that quantifies the relationship between two subspaces, representing the rotational transformation from one subspace to another. The dimensions of subspaces F and G can differ but are both fixed to q in the present context. The principal angles $\{\theta_j\}_{j=1}^q$ (also known as subspace angles) and the associated principal vectors $\{f_j, g_j\}_{j=1}^q$ of the two subspaces are defined recursively as [28,46]:

$$\cos \theta_j = \langle f_j, g_j \rangle = \max_{\substack{f \in F, \|f\|_2=1 \\ f \perp f_1, \dots, f_{j-1}}} \max_{\substack{g \in G, \|g\|_2=1 \\ g \perp g_1, \dots, g_{j-1}}} |f^* g|, \quad (14)$$

where the principal angles $\theta_j \in [0, \pi/2]$ for $j = 1, 2, \dots, q$.

Subspace angle interpolation (SAI) generates an intermediate subspace from two given subspaces. As a generalization of the real-valued SAI in Lieu and Lesoinne [29], Lieu and Farhat [30], the complex-valued SAI is developed for complex subspaces. These subspaces are associated with DMD modes corresponding to different parameters of a dynamical system. The intermediate subspace, in turn, is associated with a target parameter that lies between the parameters of the two given subspaces. Specifically, the subspaces F and G are spanned by the linearly independent columns of matrices Φ_1 and Φ_2 , which represent the DMD modes at two distinct system parameters, γ_1 and γ_2 , respectively. As previously illustrated, $\hat{\Phi}_j$ denotes the orthonormalized matrices obtained from the DMD modes Φ_j . Consequently, the principal angles and vectors are then computed via SVD:

$$LQR^* = \hat{\Phi}_1^* \hat{\Phi}_2 \in \mathbb{C}^{N_r \times N_r}. \quad (15)$$

The corresponding principal angles are computed as:

$$\Theta = \cos^{-1} Q \in \mathbb{R}^{N_r \times N_r}, \quad (16)$$

where the diagonal entries of the matrix Q represent the singular values of the matrix $\hat{\Phi}_1^* \hat{\Phi}_2$. The principal vectors are the columns of matrices:

$$F = \hat{\Phi}_1 L \text{ and } G = \hat{\Phi}_2 R. \quad (17)$$

The matrix $\bar{\Theta}$, which contains the interpolated principal angles, is computed via linear interpolation:

$$\bar{\Theta} = \left(\frac{\bar{\gamma} - \gamma_1}{\gamma_2 - \gamma_1} \right) \Theta, \quad \bar{\gamma} \in [\gamma_1, \gamma_2], \quad (18)$$

where $\bar{\gamma}, \gamma_1$, and γ_2 are parameters of the dynamical system.

The DMD modes $\bar{\Phi}$ at the target $\bar{\gamma}$ are obtained from the interpolated intermediate subspace:

$$\bar{\Phi} = \left(F \frac{\sin(\Theta - \bar{\Theta})}{\sin \Theta} + G \frac{\sin \bar{\Theta}}{\sin \Theta} \right) R^*. \quad (19)$$

SAI is computationally efficient and suitable for generating an intermediate subspace between two subspaces. However, extending this method to interpolate multiple subspaces with higher accuracy is challenging.

3.5. Grassmann manifold interpolation

Grassmann manifold interpolation (GMI) provides a geometrically consistent framework for interpolating subspaces. While the detailed derivation of real-valued GMI is well-established in prior work [23,31,47], this framework cannot accommodate the complex-valued and non-orthonormal nature of DMD modes. Here, the GMI framework is extended to the complex Grassmann manifold through the compact Stiefel manifold representation, thereby constructing a complex-valued GMI framework. Specifically, the geodesic, exponential mapping, and logarithmic mapping on the complex Grassmann manifold—represented through the Stiefel manifold—are derived in detail in the [Appendix A](#).

The DMD modes $\{\Phi_j\}_{j=1}^{N_p}$, obtained via standard DMD, are associated with a set of parameters $\{\gamma_j\}_{j=1}^{N_p}$ of the dynamical system. These matrices are recognized as spanning subspaces $\{\mathcal{Y}_j\}_{j=1}^{N_p}$ on the Grassmann manifold $\mathcal{G}$. Consequently, a subspace $\bar{\mathcal{Y}}$ spanned by

the DMD modes $\bar{\Phi}$ at the target $\bar{\gamma}$ can be constructed from the known subspaces $\{\mathcal{Y}_j\}_{j=1}^{N_p}$. Initially, a reference point $\mathcal{Y}_0$ is selected from the subspaces $\{\mathcal{Y}_j\}_{j=1}^{N_p}$ to serve as the origin for the tangent space $\mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ of the Grassmann manifold $\mathcal{G}$. Then, the points $\{\mathcal{Y}_j\}_{j=1}^{N_p}$ are projected onto the tangent space $\mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$. This projection is realized via the logarithmic mapping $\text{Log}_{\mathcal{Y}_0} : \mathcal{G} \rightarrow \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$, defined as:

$$(I - \hat{\Phi}_0 \hat{\Phi}_0^*) \hat{\Phi}_j (\hat{\Phi}_0^* \hat{\Phi}_j)^{-1} = M_j \hat{\Xi}_j N_j^* \quad (\text{thin SVD}), \quad (20a)$$

$$\Gamma_j = M_j \tan^{-1} \hat{\Xi}_j N_j^*, \quad (20b)$$

where $\hat{\Phi}_j$ are the orthonormalized DMD modes, Γ_j is the matrix representation of the projection $\mathcal{X}_j = \text{Log}_{\mathcal{Y}_0} \mathcal{Y}_j \in \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ for $\mathcal{Y}_j \in \mathcal{G}$, with $\text{Log}_{\mathcal{Y}_0} \mathcal{Y}_0 = 0$. Each projection in $\{\mathcal{X}_j\}_{j=1}^{N_p} \in \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ denotes an initial derivative, which can uniquely determine a geodesic (i.e., the shortest path joining any two points on the manifold) on the manifold in combination with the initial position defined by the reference point $\mathcal{Y}_0$. This means that all such geodesics share the same initial position, and the interpolation constructs an initial derivative $\bar{\mathcal{X}} \in \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ related to the target parameter $\bar{\gamma}$, which determines a geodesic on the manifold $\mathcal{G}$ for $\bar{\gamma}$ [48]. Importantly, this process requires that $\{\mathcal{Y}_j\}_{j=1}^{N_p}$ lie within a sufficiently close neighborhood of the reference point $\mathcal{Y}_0$ on the Grassmann manifold $\mathcal{G}$, as determined by the distance metric discussed in Section 3.3.

As discussed in the Appendix A, the tangent space $\mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ is a vector space, enabling the straightforward application of Euclidean interpolation algorithms. The initial derivative $\bar{\mathcal{X}} \in \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ of a geodesic for $\bar{\gamma}$ that satisfies $\bar{\mathcal{X}} = \text{Log}_{\mathcal{Y}_0} \bar{\mathcal{Y}}$ is represented by the matrix $\bar{\Gamma}$, which is computed as:

$$\bar{\Gamma} = \sum_{j=1}^{N_p} \sigma_j \Gamma_j, \quad (21)$$

where $\{\sigma_j\}$ are the interpolation weights derived from Lagrange polynomials.

Once $\bar{\mathcal{X}} \in \mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ is determined, it can be mapped back to the manifold $\mathcal{G}$, yielding the endpoint of the geodesic corresponding to $\bar{\gamma}$. This mapping is implemented via the exponential mapping $\text{Exp}_{\mathcal{Y}_0} \mathcal{G} : \mathcal{T}_{\mathcal{Y}_0}\mathcal{G} \mapsto \mathcal{G}$, which transforms points in the tangent space $\mathcal{T}_{\mathcal{Y}_0}\mathcal{G}$ back onto the manifold $\mathcal{G}$. It is given explicitly by:

$$\bar{\Gamma} = \bar{M} \bar{\Xi} \bar{N}^* \quad (\text{thin SVD}), \quad (22a)$$

$$\bar{\Phi} = (\hat{\Phi}_0 \bar{N} \cos \bar{\Xi} + \bar{M} \sin \bar{\Xi}) \bar{N}^*, \quad (22b)$$

where $\bar{\Phi}$ is column-orthonormal and represents the subspace $\bar{\mathcal{Y}} \in \mathcal{G}$ at $\bar{\gamma}$.

3.6. Pointwise interpolation and mode-realigned pointwise interpolation

The idea of incorporating pointwise interpolation (PWI) into DMD framework was proposed in Huhn et al. [21], which applied Lagrange interpolation to either the low-dimensional linear operator or its eigenvalues and eigenvectors to construct PROMs. Here, this principle is extended to DMD modes. As previously described, the standard DMD provides the DMD modes $\{\Phi_j\}_{j=1}^{N_p}$, which correspond to the parameter set $\{\gamma_j\}_{j=1}^{N_p}$ of the dynamical system, with a group of reference modes denoted as Φ_0 . Subsequently, the DMD modes $\bar{\Phi}$ at the target $\bar{\gamma}$ are interpolated from the nearest known DMD modes $\{\Phi_j\}_{j=1}^{N_p}$. To clarify, this can be expressed as:

$$\bar{\Phi} = \sum_{j=1}^{N_p} \sigma_j \Phi_j, \quad (23)$$

where σ_j represents the interpolation functions, consistent with Eq. (21).

As previously mentioned, DMD modes are complex vectors, which causes subtleties in defining angles. In this context, the concept of the (Kasner's) pseudo-angle is introduced to measure the relationships between complex vectors. Specifically, for two complex vectors, ϕ_a and ϕ_b , the pseudo-angle φ is defined as follows [49–51]:

$$e^{i\varphi(\phi_a, \phi_b)} = \frac{\langle \phi_a, \phi_b \rangle}{|\langle \phi_a, \phi_b \rangle|}. \quad (24)$$

This angle can be directly calculated via $\varphi = -i \ln e^{i\varphi}$ by requiring $\varphi \in [-\pi, \pi]$. Here, the complex inner product is given by $\langle \phi_a, \phi_b \rangle = \phi_a^* \phi_b$.

Building on this concept, we propose mode-realigned pointwise interpolation (MRPWI), a novel method that integrates the pseudo-angle with PWI. It is worth mentioning that the DMD modes are the columns of $\Phi_j = [\phi_j^1, \phi_j^2, \dots, \phi_j^{N_r}]$, where $\Phi_j \in \mathbb{C}^{N_n \times N_r}$ and each $\phi_j^k \in \mathbb{C}^{N_n}$ for $k = 1, 2, \dots, N_r$. Once the reference modes $\Phi_0 = [\phi_0^1, \phi_0^2, \dots, \phi_0^{N_r}]$ are selected, the modes $\{\Phi_j\}_{j=1}^{N_p}$ are rotated columnwise to align with Φ_0 . The realignment process is defined as:

$$\phi_j^k \rightarrow \phi_j^k e^{-i\varphi_j^k}, \text{ for } j = 1, \dots, N_p, k = 1, 2, \dots, N_r, \quad (25)$$

where i is the imaginary unit, and φ_j^k is the pseudo-angle $\varphi(\phi_0^k, \phi_j^k)$ between ϕ_0^k and ϕ_j^k . Finally, the DMD modes $\bar{\Phi}$ at $\bar{\gamma}$ are computed from the realigned DMD modes $\{\Phi_j\}_{j=1}^{N_p}$, as described in Eq. (23).

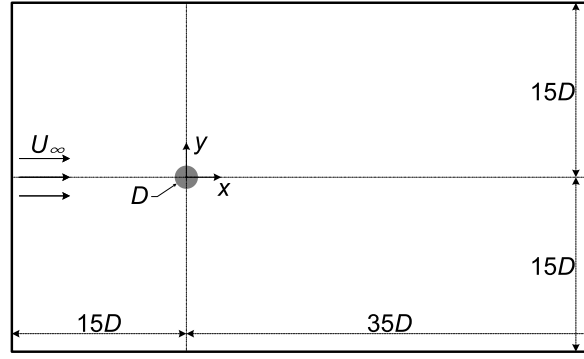


Fig. 1. Computational domain configuration for flow over a cylinder, with the cylinder diameter doubled for visualization clarity.

The effectiveness of MRPWI can be initially elucidated from a physical perspective. For conventional ROMs, applying an arbitrary phase shift θ (i.e., multiplying by $e^{i\theta}$) to any DMD mode would not alter its essence. As a result of this phase freedom, applying PWI to DMD modes may introduce significant errors even when system parameters are sufficiently close. For instance, when $\gamma_1 = \bar{\gamma} - \epsilon$ and $\gamma_2 = \bar{\gamma} + \epsilon$ with $\epsilon \rightarrow 0^+$, one would reasonably expect that interpolating the DMD modes (after mode ranking) of γ_1 and γ_2 at $\bar{\gamma}$ should yield a PROM fully consistent with the ROM constructed directly at $\bar{\gamma}$. However, phase freedom permits $\phi_2^k = \phi_1^k e^{i\pi} = -\phi_1^k$, leading to the degeneracy of $(\phi_1^k + \phi_2^k)/2$. From this perspective, mode realignment (with either $\gamma_0 = \gamma_1$ or $\gamma_0 = \gamma_2$) eliminates such phase freedom, ensuring that modes derived from identical systems become strictly identical. Therefore, the PROMs constructed via MRPWI of $\bar{\gamma} \pm \epsilon$ converge to the ROMs at $\bar{\gamma}$. Furthermore, the theoretical foundation and geometric interpretation of mode realignment are detailed in [Appendix B](#).

4. Test case: flow over a cylinder

PROMs are demonstrated and assessed through the two-dimensional flow over a cylinder [52]. The flow behavior depends only on the Reynolds number, ranging from creeping flows ($Re \rightarrow 0$) to fully turbulent flows ($Re \approx 10^7$) as the Reynolds number increases [53]. The steady symmetric flow undergoes a Hopf bifurcation at $Re \approx 47$, transitioning to periodic vortex shedding. This unsteady flow remains two-dimensional laminar until $Re \approx 200$ [54]. As the Reynolds number increases above $Re \approx 200$, the flow becomes three-dimensional while preserving strong two-dimensional coherence, corresponding to a second bifurcation.

4.1. Case description and numerical methods

As sketched in [Fig. 1](#), the coordinate origin is located at the cylinder's center with diameter D , while the computational domain is $[-15D, 35D] \times [-15D, 15D] \setminus \{(x, y) | \sqrt{x^2 + y^2} < D/2\}$. $\mathbf{u} = (u, v)^T$ and p represent the velocity and pressure, respectively. A Dirichlet boundary condition with $\mathbf{u} = U_\infty \mathbf{e}_x$ is imposed at the inlet $x = -15D$. At the outlet ($x = 35D$), a natural boundary condition, expressed as $[-p\mathbf{I} + \nu \nabla \mathbf{u}] \cdot \mathbf{n} = 0$ with $\mathbf{n}$ as the normal vector, is applied [55]. Additionally, no-slip wall boundary conditions are enforced on the cylinder wall. Periodic boundary conditions are applied at the top and bottom boundaries ($y = \pm 15D$) to emulate an unbounded flow domain and eliminate wall interference [56]. The initial velocity field is set as $\mathbf{u} = U_\infty \mathbf{e}_x$, and asymmetry naturally emerges in the computed flow field. The governing equations are the unsteady, two-dimensional, incompressible Navier-Stokes equations:

$$\nabla \cdot \mathbf{u} = 0, \quad (26a)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u}, \quad (26b)$$

with all variables non-dimensionalized using D , U_∞ and constant density ρ . The Reynolds number is defined as $Re = U_\infty D / \nu$ where ν is the kinematic viscosity.

The open-source code Nek5000 numerically integrates the unsteady two-dimensional incompressible Navier-Stokes equations utilizing the high-order spectral element method (SEM) [57]. This method combines the high accuracy of spectral methods with the geometrical flexibility of the finite element method, offering advantages such as low numerical dispersion and dissipation [58]. The present study adopts the $\mathbb{P}_N - \mathbb{P}_{N-2}$ formulation, which expands the solution in terms of Lagrange polynomials of order not larger than N along each spatial direction within a spectral element. These polynomials are evaluated at $N + 1$ Gauss-Lobatto-Legendre (GLL) points (the zeros of the first derivative of the N^{th} -order Legendre polynomial and the endpoints) for velocity and at $N - 1$ Gauss-Legendre (GL) points (the zeros of the $(N + 1)^{\text{th}}$ -order Legendre polynomial) for pressure [55,59,60]. In other words, the velocity and pressure are defined on the GLL mesh and the GL mesh, respectively, ensuring that the velocity field is continuous across element boundaries. The present computational domain is discretized with 1472 spectral elements, each containing $8 \times 8 = 64$ GLL quadrature points, which correspond to a maximum polynomial order of $N = 7$. Body-fitted elements are employed, with local refinement near the cylinder wall and in the wake region to accurately capture boundary layer effects and wake dynamics. The third-order backward

difference formula/extrapolation (BDF3/EXT3) is employed for time integration. Specifically, the viscous terms are treated implicitly via BDF3, while the convective terms are integrated explicitly through EXT3 [61]. The fixed time step is set to $\Delta\tau = 0.001$.

Direct numerical simulations (DNS) are performed for $Re \in \{70, 80, 85, 100, 110, 115, 120, 125, 130, 135, 140, 145, 150, 160, 175, 180, 190\}$, with snapshots sampled during the periodic vortex shedding regime at intervals of $\Delta t = 100\Delta\tau = 0.1$. The flow data at the target Reynolds number $\bar{Re} = 130$ are computed as a reference, and data at other Reynolds numbers are utilized to construct various PROMs for $\bar{Re} = 130$. These PROMs are then analyzed and evaluated against the reference data. Initially, the significance of mode ranking and mode realignment for the construction of PROMs is thoroughly examined. Subsequently, the accuracy and sensitivity of PROMs constructed with SAI, GMI, and PWI methods are investigated with respect to the number of modes N_r (i.e., truncation), the system parameter interval ΔRe , and the number of neighbors N_p . Finally, the PROMs generated through various interpolation methods are evaluated for specific parameter configurations, with a focus on flow fields and aerodynamic coefficients.

4.2. Importance of mode ranking and mode realignment

Essentially, both mode ranking and realignment aim to improve the similarity of the counterparts for interpolation in the construction of PROMs, specifically the features of DMD modes and eigenvalues at different Reynolds numbers.

Although mode ranking may appear trivial, it is essential and effective in the construction of PROMs. As previously mentioned, DMD modes and eigenvalues need to be ranked according to the criterion outlined in Section 3.2. The continuous-time DMD eigenvalues occur in complex conjugate pairs, resulting in a distribution symmetric about the real axis (i.e., $\text{Im}(s_j) = 0$), as depicted in Fig. 2. Furthermore, these eigenvalues approximately lie along the imaginary axis (i.e., $\text{Re}(s_j) \approx 0$), reflecting purely oscillatory dynamics

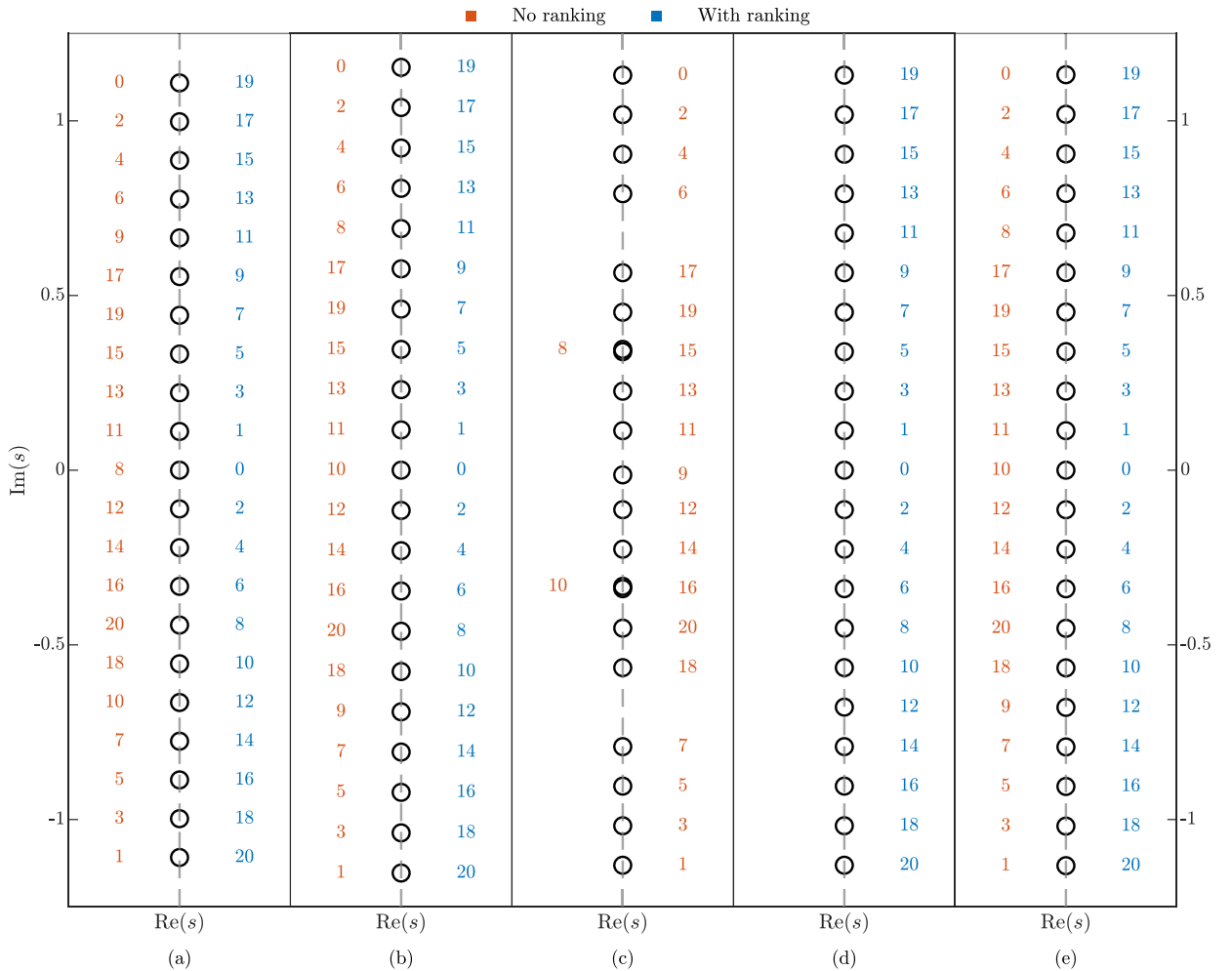


Fig. 2. DMD eigenvalues obtained through standard DMD or interpolation: (a) Standard DMD eigenvalues at $Re = 120$ without and with ranking, (b) Standard DMD eigenvalues at $Re = 140$ without and with ranking, (c) DMD eigenvalues via interpolation at $Re = 130$ without ranking, (d) DMD eigenvalues via interpolation at $Re = 130$ with ranking, (e) Standard DMD eigenvalues at $Re = 130$ without and with ranking.

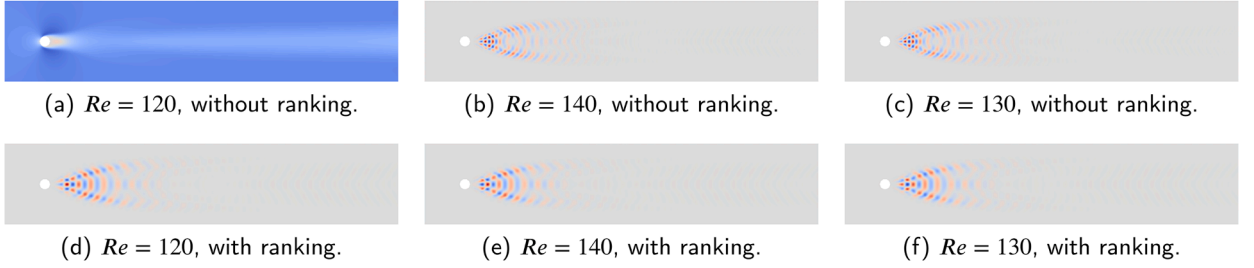


Fig. 3. DMD mode 8 (streamwise velocity component), with and without ranking.

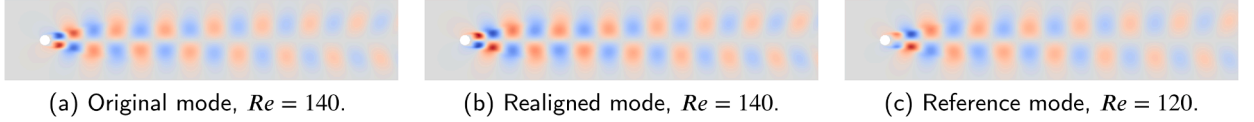


Fig. 4. DMD mode 1 (streamwise velocity component), with ranking: original, realigned, and reference modes.

without exponential growth or decay. This behavior corresponds to the periodic vortex shedding characteristic of the flow over a cylinder. Additionally, the eigenvalue at the origin ($\text{Re}(s_j) \approx 0$ and $\text{Im}(s_j) \approx 0$) corresponds to the mean flow (i.e., time-averaged flow), representing the absence of both exponential growth/decay and oscillatory behavior.

The eigenvalues in Fig. 2 are either calculated directly using standard DMD or obtained via 2-point Lagrange interpolation. Specifically, the eigenvalues at $\text{Re} \in \{120, 140\}$, computed by standard DMD, are utilized to interpolate the corresponding eigenvalues at the target Reynolds number $\text{Re} = 130$, which are also computed using standard DMD for comparison. These eigenvalues are numbered for reference: the numbers on the left and right in Fig. 2(a,b,e) correspond to results without and with mode ranking, respectively. Similarly, the interpolated results of the eigenvalues without and with mode ranking are presented in Fig. 2(c,d), respectively. Without mode ranking, the eigenvalues computed by standard DMD lack organization, resulting in significantly disorganized and incorrect interpolated eigenvalues, as illustrated in Fig. 2(c). These discrepancies are evident when comparing the eigenvalues obtained through standard DMD with those generated via Lagrange interpolation, as shown in Fig. 2(c,e). Conversely, when the eigenvalues are ranked, the interpolated eigenvalues closely align with those from standard DMD, as depicted in Fig. 2(d,e).

In particular, it can be seen that the interpolated eigenvalues associated with DMD mode 8 and mode 10 are incorrect in Fig. 2(c). Therefore, the following discussion focuses on these DMD modes at different Reynolds numbers discussed previously. Briefly, only mode 8 (streamwise velocity component) is presented in Fig. 3, showing the real part, since the real and imaginary components of the complex-valued DMD modes exhibit similar patterns. Clearly, the corresponding DMD mode in Fig. 3(c) cannot be obtained by interpolating the DMD modes in Fig. 3(a,b) without mode ranking. By contrast, when mode ranking is applied, the DMD modes in Fig. 3(d,e) share similar spatial structures, making them appropriate for interpolation, as demonstrated in Fig. 3(f).

The efficacy of mode realignment is demonstrated by DMD mode 1 (with ranking) in Fig. 4. We align the mode at $\text{Re} = 140$ (Fig. 4(a)) with its counterpart at $\text{Re} = 120$ (Fig. 4(c)). As shown, the modes become more similar after alignment (Fig. 4(b,c)), making them more suitable for interpolation and achieving the desired results.

4.3. Comparison of PROMs with different interpolation methods

Upon construction of the PROMs at the target parameter, the relative L^2 errors (RLEs) are introduced to evaluate their accuracy, defined as:

$$\bar{\varepsilon} = \sqrt{\frac{\sum_{j=1}^{N_r} \|\mathbf{x}_j^{\text{ROM/PROM}} - \mathbf{x}_j^{\text{DNS}}\|_2^2}{\sum_{j=1}^{N_r} \|\mathbf{x}_j^{\text{DNS}}\|_2^2}}. \quad (27)$$

Here, $\mathbf{x}_j$ denotes the snapshots obtained from either DNS, an ROM, or a PROM, where the DNS flow data serves as the reference. The total number of snapshots used for averaging is denoted by N_r .

The PROMs at the target Reynolds number $\text{Re} = 130$ are constructed via SAI, GMI, or MRPWI, with configurations including the number of modes N_r , the system parameter interval ΔRe , and the number of neighbors N_p . The RLEs of velocity and pressure for varying N_r are presented in Fig. 5. The 2-point and 4-point interpolation (i.e., $N_p \in \{2, 4\}$) correspond to $\text{Re} \in \{120, 140\}$ and $\text{Re} \in \{100, 120, 140, 160\}$, respectively, with $\Delta \text{Re} = 20$. The RLEs consistently decrease for both the ROMs and PROMs, stabilizing beyond certain values of N_r . Fig. 6 demonstrates the relationship between the RLEs and ΔRe . The Reynolds numbers for 2-point and 4-point interpolation (i.e., $N_p \in \{2, 4\}$) are $\text{Re} \in \{125, 135\}, \{120, 140\}, \{115, 145\}, \{110, 150\}$ and $\text{Re} \in \{115, 125, 135, 145\}, \{100, 120, 140, 160\}, \{85, 115, 145, 175\}, \{70, 110, 150, 190\}$, respectively, corresponding to $\Delta \text{Re} \in \{10, 20, 30, 40\}$, with $N_r = 11$. The RLEs increase monotonically with increasing ΔRe . The influence of N_p is illustrated in Fig. 7, where N_p varies

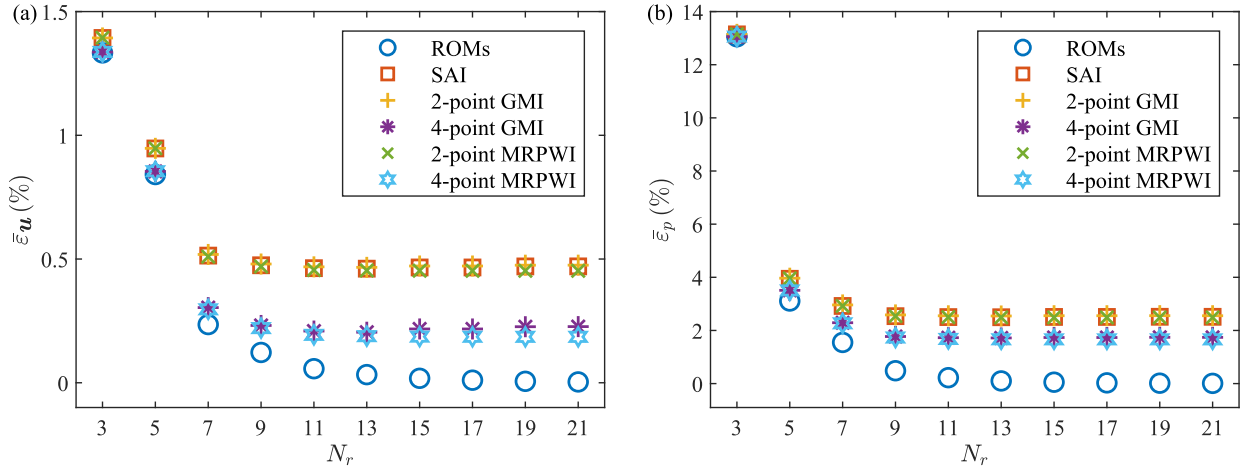


Fig. 5. RLEs versus number of modes N_r : (a) velocity, and (b) pressure. (2-point: $Re \in \{120, 140\}$; 4-point: $Re \in \{100, 120, 140, 160\}$).

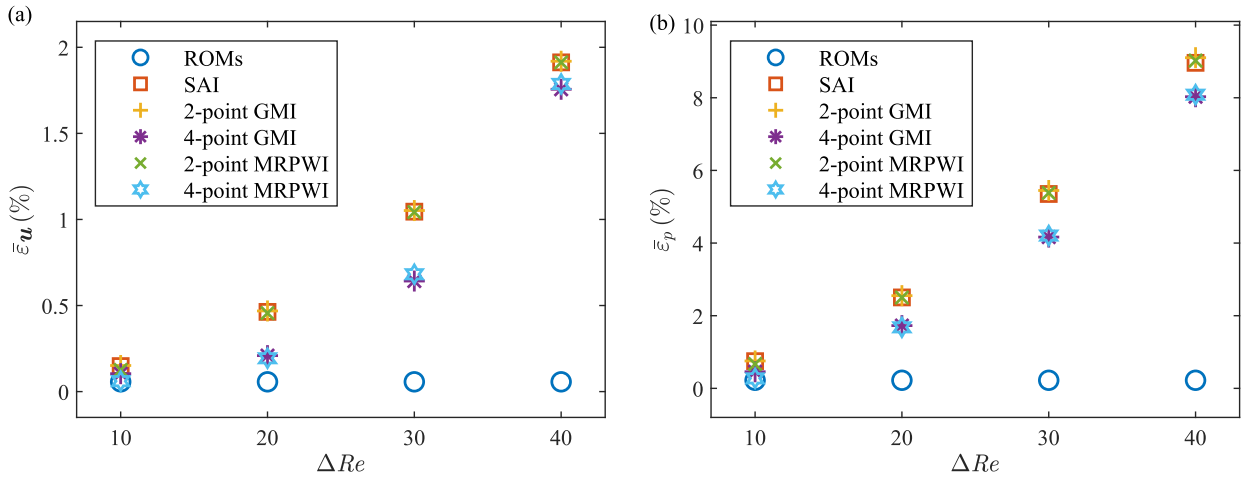


Fig. 6. RLEs versus system parameter interval ΔRe : (a) velocity, and (b) pressure. (2-point: $Re \in \{125, 135\}, \{120, 140\}, \{115, 145\}, \{110, 150\}$; 4-point: $Re \in \{115, 125, 135, 145\}, \{100, 120, 140, 160\}, \{85, 115, 145, 175\}, \{70, 110, 150, 190\}$).

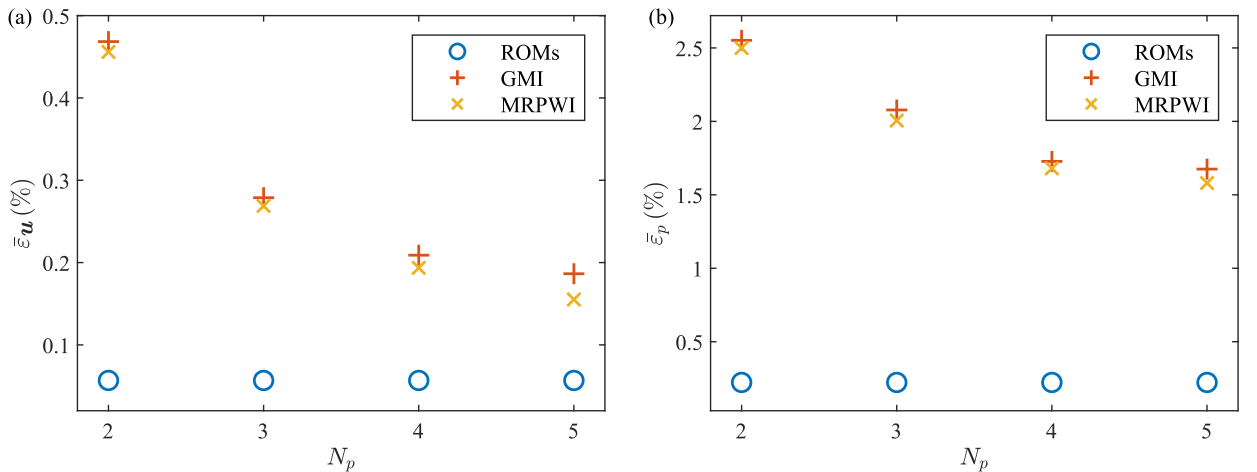


Fig. 7. RLEs versus number of neighbors N_p : (a) velocity, and (b) pressure. ($Re \in \{120, 140\}, \{100, 120, 140\}, \{100, 120, 140, 160\}, \{80, 100, 120, 140, 160\}$).

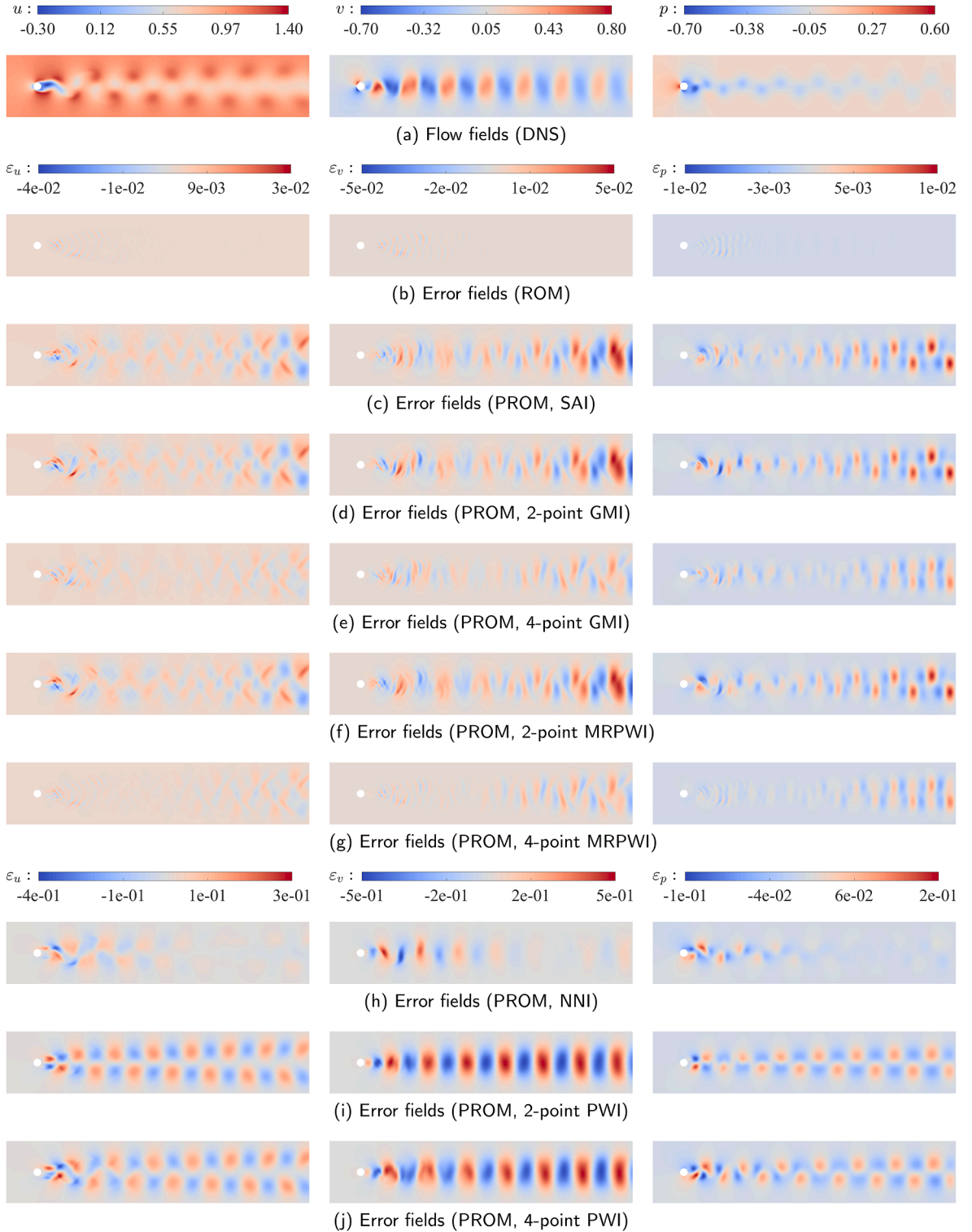


Fig. 8. Flow fields (DNS) and error fields (ROM/PROMs versus DNS) at $\overline{Re} = 130$, $t = 15$. Parameter configurations: $N_r = 11$, $\Delta Re = 20$, $N_p \in \{2, 4\}$. Neighboring cases: NNI ($Re = 140$), 2-point ($Re \in \{120, 140\}$), 4-point ($Re \in \{100, 120, 140, 160\}$).

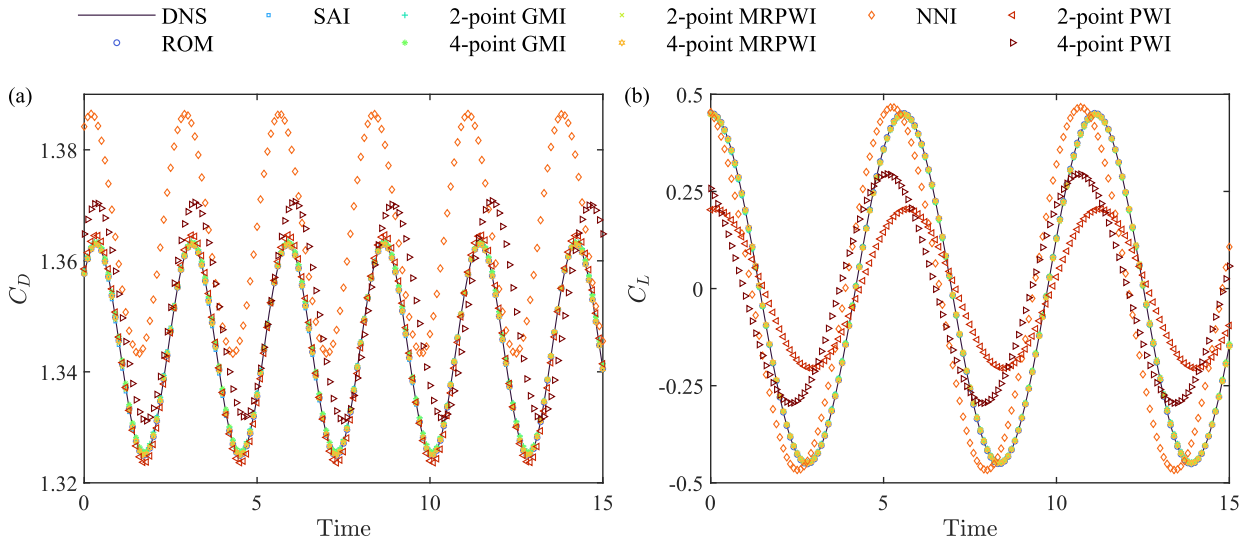


Fig. 9. Time evolution of (a) drag coefficient and (b) lift coefficient at $\overline{Re} = 130$. Parameter configurations: $N_r = 11$, $\Delta Re = 20$, $N_p \in \{2, 4\}$. Neighboring cases: NNI ($Re = 140$), 2-point ($Re \in \{120, 140\}$), 4-point ($Re \in \{100, 120, 140, 160\}$).

as $Re \in \{120, 140\}, \{100, 120, 140\}, \{100, 120, 140, 160\}, \{80, 100, 120, 140, 160\}$ (i.e., $N_p \in \{2, 3, 4, 5\}$), with $N_r = 11$ and $\Delta Re = 20$. The trend in RLEs demonstrates that higher-order interpolation, which incorporates more neighboring cases, yields reduced RLE values.

In general, the accuracy of PROMs improves with an increase in N_r and N_p , whereas it deteriorates with increasing ΔRe . The PROMs generated using 2-point GMI or 2-point MRPWI demonstrate performance comparable to those with SAI. Regardless of N_r , ΔRe , or N_p , the accuracy of the PROMs produced by MRPWI and GMI shows no significant difference. Representative intermediate values, $N_r = 11$, $\Delta Re = 20$, and $N_p \in \{2, 4\}$, corresponding to $Re \in \{120, 140\}$ and $\{100, 120, 140, 160\}$, respectively, are selected for further demonstration and analysis.

Fig. 8 presents the velocity fields $\mathbf{u} = (u, v)^T$, pressure field p , and their error fields corresponding to $\overline{Re} = 130$ at $t = 15$. Specifically, the first row depicts the flow fields obtained by DNS (Fig. 8(a)), with the remaining rows presenting error fields derived from the ROM and PROMs (Fig. 8(b-j)). Here, the error fields ε_j are defined by the discrepancies $(\mathbf{x}_j^{\text{ROM/PROM}} - \mathbf{x}_j^{\text{DNS}})$ between the ROM/PROM solution and the DNS reference, scaled by the relevant reference quantities. Overall, the flow fields from the PROMs constructed by SAI, GMI, and MRPWI closely match those from DNS. These PROMs accurately capture the flow features, with RLEs below 0.5 % for velocity fields and 2.6 % for pressure fields. Both 2-point interpolation methods (SAI, GMI, MRPWI) and 4-point interpolation methods (GMI, MRPWI) demonstrate highly identical error patterns and magnitudes within their respective groups. Notably, the 4-point methods achieve significantly lower error magnitudes compared to their 2-point counterparts. A comparison between high-amplitude regions in DNS fields and error fields reveals that amplitude discrepancy constitutes a dominant error component. Meanwhile, small-scale oscillatory patterns are observed, mainly in regions either close to or distant from the cylinder. Additionally, the PROMs generated via nearest-neighbor interpolation (NNI) and PWI are included for comparison (Fig. 8(h-j)). Here, NNI is implemented by directly using the nearest neighboring case at $Re = 140$ without performing any interpolation. NNI and PWI exhibit significantly larger error magnitudes, exceeding those of the proposed interpolation methods (SAI, GMI, and MRPWI) by approximately one order of magnitude. For these PROMs, the RLEs for velocity and pressure are 2.3 % and 12.7 % with NNI, 8.4 % and 30.2 % with 2-point PWI, as well as 7.9 % and 30.4 % with 4-point PWI. Clearly, the PROMs created with NNI and PWI fail to accurately capture the features of nonlinear complex flows in the wake region. This indicates that the PROMs built by the innovative interpolation methods exhibit significant improvements in accuracy.

The aerodynamic forces acting on the cylinder wall, including lift and drag, are derived from pressure and viscous forces. These forces are obtained through integration along the cylinder surface. The drag and lift coefficients are defined as:

$$C_D = \frac{F_x}{\rho U_\infty^2 D/2}, \quad C_L = \frac{F_y}{\rho U_\infty^2 D/2}, \quad \mathbf{F} = \oint_{\partial\Omega} (-p\mathbf{I} + \boldsymbol{\tau}) \cdot \mathbf{n} dS. \quad (28)$$

Here, ρ is the fluid density, U_∞ is the free-stream velocity, and D is the cylinder diameter. The force vector $\mathbf{F} = (F_x, F_y)$ is calculated by the pressure p , the viscous stress tensor $\boldsymbol{\tau}$, and the unit normal vector $\mathbf{n}$ pointing outward from the cylinder surface $\partial\Omega$. As shown in Fig. 9, the aerodynamic coefficients computed by PROMs constructed with SAI, GMI, and MRPWI demonstrate excellent agreement with those from DNS and the ROM. By contrast, the PROMs generated via NNI and PWI fail to achieve comparable accuracy in predicting these coefficients.

Table 1
Computational time for PROMs constructed via the proposed interpolation methods.

-	SAI	GMI	MRPWI
2-point interpolation	0.59 s	1.05 s	0.22 s
4-point interpolation	-	1.91 s	0.36 s

4.4. Computational complexity

The efficiency of PROMs constructed by the proposed interpolation methods, including SAI, GMI, and MRPWI, is evaluated based on computational time, as shown in Table 1. In general, MRPWI generates the most efficient PROMs, with computational speeds approximately 2.5 times and 5 times faster than those obtained from SAI and GMI, respectively.

Additionally, the total computational complexity of SAI, GMI, and MRPWI is $\mathcal{O}(N_n N_r^2)$, $\mathcal{O}(N_p N_n^2 N_r)$, and $\mathcal{O}(N_p N_n N_r)$, respectively, given that $N_n \gg N_r$. This indicates that MRPWI achieves the lowest computational complexity among the three methods.

5. Conclusions

Novel data-driven DMD PROMs are constructed through the developed interpolation methods in conjunction with the DMD framework. These interpolation methods, including SAI, GMI and MRPWI, are employed to interpolate complex-valued, non-orthonormal DMD modes, while eigenvalues are uniquely determined via Lagrange interpolation. Significantly, we demonstrate that mode ranking is crucial in combination with these interpolation methods, particularly when a large number of DMD modes are involved. These PROMs are assessed through the numerical case of two-dimensional flow over a cylinder. By examining the evolution of flow fields and aerodynamic coefficients obtained from DNS, ROMs, and PROMs, it is evident that the generated PROMs are highly accurate. The accuracy of these PROMs improves with increasing number of modes N_r , increasing number of neighbors N_p , and decreasing system parameter interval ΔRe . Specifically, 2-point interpolation yields PROMs with nearly equivalent accuracy regardless of whether SAI, GMI, or MRPWI is employed. Moreover, the PROMs developed with MRPWI demonstrate accuracy comparable to those constructed with GMI across various configurations (N_r , N_p and ΔRe). Notably, MRPWI combines theoretical simplicity with computational efficiency for the construction of PROMs, demonstrating broad applicability and robust performance.

Although the PROMs demonstrate significant accuracy for the present single-parameter problem, the proposed interpolation methods still require adaptation for multi-parametric problems. It is also worth mentioning that the proposed interpolation methods could be extended to the DMD-Galerkin framework for constructing PROMs, though their efficacy remains to be assessed.

CRedit authorship contribution statement

Lei Du: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Shengqi Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization; **Ruili Zhang:** Methodology, Formal analysis, Conceptualization; **Shibin Zhang:** Methodology, Formal analysis, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Derivations of exponential and logarithmic mappings on complex Grassmann manifold via the compact Stiefel manifold representation

Inspired by the derivations of exponential and logarithmic mappings on the real Grassmann manifold [37,44,62], the complex counterparts of these mappings are derived on the complex Grassmann manifold with the compact Stiefel manifold representation in this study.

The Grassmann manifold (or Grassmannian) $\mathcal{G} = Gr(p, n)$ is a smooth manifold, locally homeomorphic to Euclidean space, and possesses a globally defined differential structure. It is defined as the set of all p -dimensional linear subspaces of the Euclidean space $\mathbb{C}^n$ for $p \leq n$ [63],

$$Gr(p, n) := \{\mathcal{Y} \subset \mathbb{C}^n | \mathcal{Y} \text{ is a subspace, } \dim(\mathcal{Y}) = p\}. \quad (\text{A.1})$$

A point $\mathcal{Y} \in \mathcal{G}$ can be represented by the column space of a matrix $\Psi \in \mathbb{C}^{n \times p}$, i.e., $\mathcal{Y} = \text{span}(\Psi)$. The matrix representation of the point $\mathcal{Y} \in \mathcal{G}$ is not unique, and infinitely many representations exist as long as their column spaces are identical.

One common representation of the Grassmann manifold is through the Stiefel manifold. Specifically, the Grassmann manifold $\mathcal{G}$ can be described as a quotient manifold of the Stiefel manifold $St(p, n)$ under the action of the unitary Lie group $U(p)$:

$$Gr(p, n) = St(p, n)/U(p) = \{[\Psi] | \Psi \in St(p, n)\}, \quad (\text{A.2})$$

where the Stiefel manifold $St(p, n)$ is defined as:

$$St(p, n) := \{\Psi \in \mathbb{C}^{n \times p} | \Psi^* \Psi = I\}, \quad (\text{A.3})$$

indicating that $\Psi \in \mathbb{C}^{n \times p}$ is a matrix with orthonormal columns. Here, $[\Psi]$ denotes the equivalence class, i.e., $[\Psi] = \{\Psi R | R \in U(p)\}$. In other words, a point on the Grassmann manifold $\mathcal{G}$ can be understood as an equivalence class containing infinitely many elements, where two matrices are equivalent if they are related by right multiplication by a unitary matrix $R \in U(p)$. Practically, if a matrix Ψ is not orthonormal, it can be orthonormalized without changing its column space.

The tangent space $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$ at a point $\mathcal{Y} \in \mathcal{G}$ is given by:

$$\mathcal{T}_{\mathcal{Y}}\mathcal{G} = \{\Gamma \in \mathbb{C}^{n \times p} | \Psi^* \Gamma = 0\}, \quad (\text{A.4})$$

where $\mathcal{Y} = [\Psi]$ is the origin of the tangent space. Having the same dimension $p(n-p)$ as $\mathcal{G}$, the tangent space $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$ can be interpreted as a local vector space approximation of the manifold at the given point $\mathcal{Y}$. Therefore, interpolation can be performed in $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$ as in Euclidean space. The Riemannian metric on the tangent space $\mathcal{T}_{\mathcal{Y}}\mathcal{G}$ can be defined as [64]:

$$\langle \Gamma_1, \Gamma_2 \rangle_{\mathcal{Y}} = \text{Re}[\text{tr}(\Gamma_1^* \Gamma_2)], \quad \Gamma_1, \Gamma_2 \in \mathcal{T}_{\mathcal{Y}}\mathcal{G}. \quad (\text{A.5})$$

Thus, geodesics on $\mathcal{G}$ are obtained by minimizing the energy functional:

$$\mathcal{E}[\Psi(t), \Psi^*(t)] = \frac{1}{2} \int_0^1 \text{tr}(\dot{\Psi}^* \dot{\Psi}) dt. \quad (\text{A.6})$$

Since $\dot{\Psi}^* \dot{\Psi}$ is Hermitian, the operation $\text{Re}[\cdot]$ is omitted. To enforce the orthonormality constraint $\Psi^* \Psi = I$, a Hermitian Lagrange multiplier $\Delta(t) \in \mathbb{C}^{p \times p}$ is introduced, leading to the augmented action:

$$S[\Psi(t), \Psi^*(t)] = \int_0^1 \left[\frac{1}{2} \text{tr}(\dot{\Psi}^* \dot{\Psi}) + \text{tr}(\Delta(\Psi^* \Psi - I)) \right] dt = \int_0^1 F(t, \Psi, \dot{\Psi}, \Psi^*, \dot{\Psi}^*) dt. \quad (\text{A.7})$$

Under the variation $\Psi(t) \rightarrow \Psi(t) + \delta\Psi(t)$ (which implies $\Psi^*(t) \rightarrow \Psi^*(t) + \delta\Psi^*(t)$ for its complex conjugate), the first variation of the action is given by:

$$\delta S = \int_0^1 \left[\left(\frac{\partial F}{\partial \Psi} - \frac{d}{dt} \frac{\partial F}{\partial \dot{\Psi}} \right) \delta\Psi + \left(\frac{\partial F}{\partial \Psi^*} - \frac{d}{dt} \frac{\partial F}{\partial \dot{\Psi}^*} \right) \delta\Psi^* \right] dt. \quad (\text{A.8})$$

The stationarity condition $\delta S = 0$ yields the Euler-Lagrange equations:

$$\ddot{\Psi} - 2\Psi\Delta = 0. \quad (\text{A.9})$$

Differentiating the constraint $\Psi^* \Psi = I$ twice with respect to t gives:

$$\ddot{\Psi}^* \Psi + 2\dot{\Psi}^* \dot{\Psi} + \Psi^* \ddot{\Psi} = 0. \quad (\text{A.10})$$

Substituting Eq. (A.9) into Eq. (A.10) and using $\Psi^* \Psi = I$ leads to:

$$\Delta = -\frac{1}{2} \dot{\Psi}^* \ddot{\Psi}. \quad (\text{A.11})$$

Thus, the geodesic equation becomes:

$$\ddot{\Psi}(t) + \Psi(t)(\dot{\Psi}^* \dot{\Psi}) = 0, \quad t \in [0, 1]. \quad (\text{A.12})$$

Differentiating $\dot{\Psi}^* \dot{\Psi}$ with respect to t gives:

$$\frac{d}{dt} \dot{\Psi}^* \dot{\Psi} = \ddot{\Psi}^* \dot{\Psi} + \dot{\Psi}^* \ddot{\Psi} = (-\Psi \dot{\Psi}^* \ddot{\Psi})^* \dot{\Psi} - \dot{\Psi}^* \Psi \ddot{\Psi} \dot{\Psi} = 0. \quad (\text{A.13})$$

In other words, the orthogonality conditions $\Psi^* \dot{\Psi} = \dot{\Psi}^* \Psi = 0$ and the geodesic equation (Eq. (A.12)), imply that $\dot{\Psi}^* \dot{\Psi}$ is constant. Given the initial condition $\mathcal{Y}(0) = [\Psi_0]$ and $\dot{\Psi}_0 = M \Xi N^*$, it follows that $\dot{\Psi}^* \dot{\Psi} = N \Xi^2 N^*$. Substituting this expression into Eq. (A.12) results in:

$$\ddot{\Psi}(t)N + \Psi(t)N\Xi^2 = 0. \quad (\text{A.14})$$

This second-order homogeneous differential equation has the solution:

$$\Psi(t)N = \Psi_0 N \cos(\Xi t) + M \sin(\Xi t). \quad (\text{A.15})$$

The geodesic is thus:

$$\mathcal{Y}(t) = \text{span}(\Psi(t)) = \text{span}(\Psi_0 N \cos(\Xi t)N^* + M \sin(\Xi t)N^*). \quad (\text{A.16})$$

Then, the exponential mapping can be defined as:

$$\Gamma = M \Xi N^* \quad (\text{thin SVD}), \quad (\text{A.17a})$$

$$\Psi = \Psi_0 N \cos \Xi N^* + M \sin \Xi N^*, \quad (\text{A.17b})$$

mapping a tangent vector to the endpoint of the geodesic that connects two points on the manifold. The equality

$$(I - \Psi_0 \Psi_0^*) \Psi (\Psi_0^* \Psi)^{-1} = M \tan \Xi N^*, \quad (\text{A.18})$$

together with the uniqueness of SVD, directly yields the logarithmic mapping (i.e., the inverse of the exponential mapping), leading to:

$$(I - \Psi_0 \Psi_0^*) \Psi (\Psi_0^* \Psi)^{-1} = M \hat{\Xi} N^* \Rightarrow \Gamma = M \tan^{-1} \hat{\Xi} N^*. \quad (\text{A.19})$$

As mentioned previously, in practical applications, non-orthonormal Ψ can be orthonormalized. The resulting expressions can still be employed, as summarized below:

$$\text{Exp}_{\mathcal{Y}_0} \mathcal{G}: \mathcal{T}_{\mathcal{Y}_0} \mathcal{G} \ni \Gamma \rightarrow \Psi \in \mathcal{G}:$$

$$\Gamma = M \Xi N^*, \quad (\text{thin SVD}) \quad (\text{A.20a})$$

$$\Psi = (\Psi_0 N \cos \Xi + M \sin \Xi) N^*. \quad (\text{A.20b})$$

$$\text{Log}_{\mathcal{Y}_0} \mathcal{G}: \mathcal{G} \ni \Psi \rightarrow \Gamma \in \mathcal{T}_{\mathcal{Y}_0} \mathcal{G}:$$

$$(I - \Psi_0 \Psi_0^*) \Psi (\Psi_0^* \Psi)^{-1} = M \hat{\Xi} N^*, \quad (\text{thin SVD}), \quad (\text{A.21a})$$

$$\Gamma = M \tan^{-1} \hat{\Xi} N^*. \quad (\text{A.21b})$$

Appendix B. Theoretical framework and geometric interpretation for mode realignment

The mathematical framework of mode realignment, encompassing both theoretical foundation and geometric interpretation, is rigorously established. Vectors $z_j \in \mathbb{C}^n$ with $j = 1, \dots, m$ represent the DMD modes of the same ranking across m cases. For the following explanations, a reference vector z_0 is chosen from these m vectors.

As discussed in Scharnhorst [51], Estrada [65], multiple angle definitions exist in complex vector spaces, including the Euclidean angle, the Hermitian angle, and the (Kasner's) pseudo-angle:

$$\text{Euclidean angle: } \cos \theta_E(z_0, z_j) = \frac{\text{Re}[\langle z_0, z_j \rangle]}{|z_0| |z_j|}, \quad \theta_E \in [0, \pi], \quad (\text{B.1a})$$

$$\text{Hermitian angle: } \cos \theta_H(z_0, z_j) = \frac{|\langle z_0, z_j \rangle|}{|z_0| |z_j|}, \quad \theta_H \in \left[0, \frac{\pi}{2}\right], \quad (\text{B.1b})$$

$$\text{(Kasner's) pseudo-angle: } e^{i\varphi(z_0, z_j)} = \frac{\langle z_0, z_j \rangle}{|\langle z_0, z_j \rangle|}, \quad \varphi \in [-\pi, \pi], \quad (\text{B.1c})$$

with the norm defined as $|z| = \sqrt{\langle z, z \rangle}$. Combining the definitions above, we obtain the relation:

$$\cos \theta_E = \cos \theta_H \cos \varphi, \quad (\text{B.2})$$

which implies $\theta_E \geq \theta_H$.

Following the conception of pseudo-angle, the realignment process performs the transformation:

$$z_j \rightarrow z'_j = z_j e^{-i\varphi(z_0, z_j)}, \quad (\text{B.3})$$

thereby aligning z_j with the reference vector z_0 . After realignment, it is straightforward to derive:

$$\langle z_0, z'_j \rangle = \langle z_0, z_j \rangle e^{-i\varphi(z_0, z_j)} = |\langle z_0, z_j \rangle| e^{i\varphi(z_0, z_j)} e^{-i\varphi(z_0, z_j)} = |\langle z_0, z_j \rangle| \in \mathbb{R}, \quad (\text{B.4})$$

and consequently

$$\varphi(z_0, z'_j) = 0, \quad \theta_E(z_0, z'_j) = \theta_H(z_0, z_j) = \theta_H(z_0, z'_j). \quad (\text{B.5})$$

These equations indicate that the mode realignment eliminates the pseudo-angle and results in the convergence of the Euclidean angle to the unchanged Hermitian angle.

For clarity, the mode realignment procedure is visualized in the decomplexification space. For a complex vector $z \in \mathbb{C}^n$, the real-valued column vector $Z \in \mathbb{R}^{2n}$ represents its decomplexification, with elements $Z_{2k-1} = \text{Re}[z_k]$ and $Z_{2k} = \text{Im}[z_k]$ for $k = 1, \dots, n$ (elementwise indexing). The phase transformation in the complex vector space and its decomplexification space can be expressed equivalently as:

$$ze^{i\phi} \sim J_\phi Z, \quad (\text{B.6})$$

where $J_\phi \in \mathbb{R}^{2n \times 2n}$ is a block-diagonal rotation matrix constructed as the direct sum:

$$J_\phi = \bigoplus_{k=1}^n \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}. \quad (\text{B.7})$$

The decomplexification vector and rotation matrix satisfy the following properties:

$$|Z| = |J_\phi Z| = |z|, \quad Z \cdot (J_{\pi/2} Z) = 0, \quad (\text{B.8})$$

and

$$Z_0 \cdot Z_j = \text{Re}[\langle z_0, z_j \rangle], \quad (J_{\pi/2} Z_0) \cdot Z_j = \text{Im}[\langle z_0, z_j \rangle]. \quad (\text{B.9})$$

The projection of Z_j onto $\text{span}\{Z_0, J_{\pi/2} Z_0\}$ is:

$$\tilde{Z}_j = (P_{Z_0} + P_{J_{\pi/2} Z_0}) Z_j = \frac{\text{Re}[\langle z_0, z_j \rangle] Z_0 + \text{Im}[\langle z_0, z_j \rangle] J_{\pi/2} Z_0}{|Z_0|^2}, \quad (\text{B.10})$$

with the projection operator defined as $P_Z = ZZ^T/|Z|^2$. Calculating the norm of $\tilde{Z}_j$ then gives:

$$|\langle z_0, z_j \rangle| = |Z_0| |\tilde{Z}_j|. \quad (\text{B.11})$$

In addition, the inner product between Z_j and any vector Z in $\text{span}\{Z_0, J_{\pi/2} Z_0\}$ equals $\tilde{Z}_j \cdot Z$.

As a result, the Euclidean angle $\theta_E(z_0, z_j)$ in the complex space is equal to its counterpart in the decomplexification space, defined by:

$$\cos \theta_E(Z_0, Z_j) = \frac{Z_0 \cdot Z_j}{|Z_0| |Z_j|}. \quad (\text{B.12})$$

The Hermitian angle can then be expressed alternatively as:

$$\cos \theta_H(z_0, z_j) = \frac{|\langle z_0, z_j \rangle|}{|z_0| |z_j|} = \frac{|\tilde{Z}_j|}{|Z_j|} = \frac{\tilde{Z}_j \cdot \tilde{Z}_j}{|Z_j| |\tilde{Z}_j|} = \frac{Z_j \cdot \tilde{Z}_j}{|Z_j| |\tilde{Z}_j|} = \cos \theta_E(Z_j, \tilde{Z}_j). \quad (\text{B.13})$$

This yields the identity: $\theta_H(z_0, z_j) = \theta_E(Z_j, \tilde{Z}_j)$. The pseudo-angle in the decomplexification space can be formulated as:

$$\cos \varphi(z_0, z_j) = \frac{\cos \theta_E(z_0, z_j)}{\cos \theta_H(z_0, z_j)} = \frac{Z_0 \cdot Z_j}{|\langle z_0, z_j \rangle|} = \frac{Z_0 \cdot \tilde{Z}_j}{|Z_0| |\tilde{Z}_j|} = \cos \theta_E(Z_0, \tilde{Z}_j), \quad (\text{B.14})$$

which means that $\varphi(z_0, z_j) = \theta_E(Z_0, \tilde{Z}_j)$.

Following the previous analysis, the realigned vector in decomplexification space admits the representation:

$$z'_j \sim Z'_j = J_{-\varphi(z_0, z_j)} Z_j. \quad (\text{B.15})$$

According to Eq. (B.4), Z'_j satisfies:

$$\cos \theta_E(J_{\pi/2} Z_0, Z'_j) = \frac{(J_{\pi/2} Z_0) \cdot Z'_j}{|J_{\pi/2} Z_0| |Z'_j|} = \frac{\text{Re}[\langle iz_0, z'_j \rangle]}{|z_0| |z_j|} = \frac{\text{Re}[-i \langle z_0, z_j \rangle]}{|z_0| |z_j|} = 0. \quad (\text{B.16})$$

Eqs. (B.5) and (B.16) establish that $\theta_E(Z_0, Z'_j) = \theta_E(z_0, z'_j) = \theta_H(z_0, z'_j) = \theta_H(z_0, z_j)$ and $\theta_E(J_{\pi/2} Z_0, Z'_j) = \pi/2$, confirming that Z'_j lies strictly within the orthogonal complement $(J_{\pi/2} Z_0)^\perp$ of $J_{\pi/2} Z_0$.

Fig. B.10 illustrates the geometric interpretation of mode realignment, comparing original modes (left panel) with realigned modes (right panel). The left panel reveals three key angular relationships for z_0 and z_j : (1) the Euclidean angle $\theta_{E,j} = \theta_E(Z_0, Z_j)$; (2) the Hermitian angle $\theta_{H,j} = \theta_H(z_0, z_j) = \theta_E(Z_j, \tilde{Z}_j)$ with $\tilde{Z}_j$ the projection of Z_j onto the Z_0 - $J_{\pi/2} Z_0$ plane; and (3) the pseudo-angle $\varphi_j = \varphi(z_0, z_j) = \theta_E(Z_0, \tilde{Z}_j)$. On the right panel, the realigned modes Z'_j lie entirely within the hyperplane $(J_{\pi/2} Z_0)^\perp$. The transformed Euclidean angle $\theta'_{E,j} = \theta_E(Z_0, Z'_j)$ and Hermitian angle $\theta'_{H,j} = \theta_H(z_0, z'_j)$ —both represented by the angle between Z_0 and Z'_j —satisfy $\theta'_{E,j} = \theta'_{H,j}$ with $\varphi(z_0, z'_j) = 0$. Consequently, in addition to eliminating phase freedom, the mode realignment also provides a meaningful geometrical property that the Euclidean and Hermitian angles coincide.

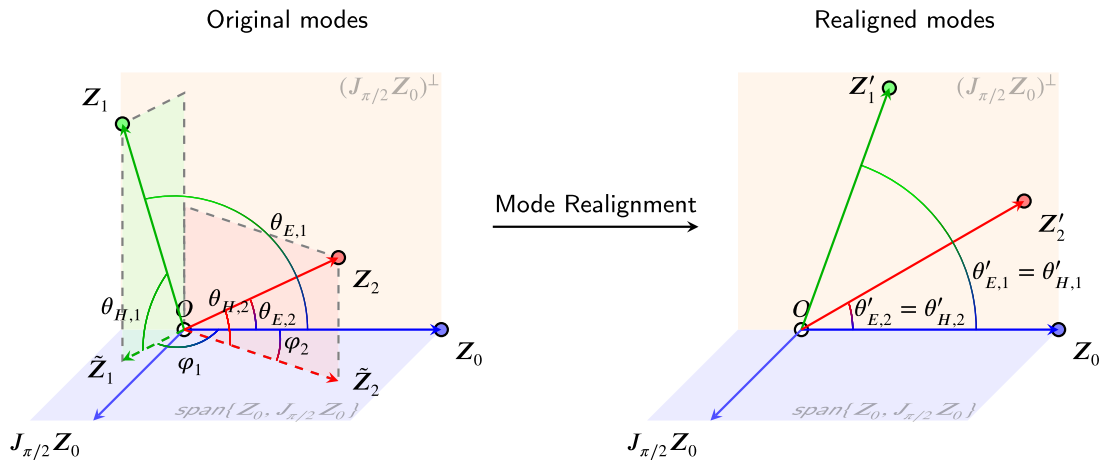


Fig. B.10. Graphical depiction of mode realignment procedure.

References

- [1] M.A. Mendez, A. Ianiro, B.R. Noack, S.L. Brunton, *Data-Driven Fluid Mechanics: Combining First Principles and Machine Learning*, Cambridge University Press, 2023.
- [2] S.L. Brunton, J.N. Kutz, *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*, Cambridge University Press, 2022.
- [3] J. Kou, W. Zhang, Data-driven modeling for unsteady aerodynamics and aeroelasticity, *Prog. Aerosp. Sci.* 125 (2021) 100725.
- [4] J.L. Lumley, *Stochastic tools in turbulence*, Academic Press, 1970.
- [5] L. Sirovich, Turbulence and the dynamics of coherent structures. I. Coherent structures, *Q. Appl. Math.* 45 (3) (1987) 561–571.
- [6] P.J. Schmid, Dynamic mode decomposition of numerical and experimental data, *J. Fluid Mech.* 656 (2010) 5–28.
- [7] A. Ehlert, C.N. Nayeri, M. Morzynski, B.R. Noack, Locally linear embedding for transient cylinder wakes, [arXiv:1906.07822](https://arxiv.org/abs/1906.07822) (2019).
- [8] E. Farzamnik, A. Ianiro, S. Discetti, N. Deng, K. Oberleithner, B.R. Noack, V. Guerrero, From snapshots to manifolds—a tale of shear flows, *J. Fluid Mech.* 955 (2023) A34.
- [9] K. Taira, M.S. Hemati, S.L. Brunton, Y. Sun, K. Duraisamy, S. Bagheri, S.T.M. Dawson, C.-A. Yeh, Modal analysis of fluid flows: applications and outlook, *AIAA J.* 58 (3) (2020) 998–1022.
- [10] C.W. Rowley, S.T.M. Dawson, Model reduction for flow analysis and control, *Annu. Rev. Fluid Mech.* 49 (1) (2017) 387–417.
- [11] J.N. Kutz, S.L. Brunton, B.W. Brunton, J.L. Proctor, *Dynamic Mode Decomposition: Data-Driven Modeling of Complex Systems*, SIAM, 2016.
- [12] C.Y. Li, Z. Chen, A.U. Weerasuriya, X. Zhang, X. Lin, L. Zhou, Y. Fu, K.T. Tim, Best practice guidelines for the dynamic mode decomposition from a wind engineering perspective, *J. Wind Eng. Ind. Aerodyn.* 241 (2023) 105506.
- [13] P. Benner, et al., *Snapshot-Based Methods and Algorithms*, Walter de Gruyter GmbH & Co KG, 2020.
- [14] C.W. Rowley, I. Mezić, S. Bagheri, P. Schlatter, D.S. Henningson, Spectral analysis of nonlinear flows, *J. Fluid Mech.* 641 (2009) 115–127.
- [15] K.K. Chen, J.H. Tu, C.W. Rowley, Variants of dynamic mode decomposition: boundary condition, Koopman, and Fourier analyses, *J. Nonlinear Sci.* 22 (2012) 887–915.
- [16] M.R. Jovanović, P.J. Schmid, J.W. Nichols, Sparsity-promoting dynamic mode decomposition, *Phys. Fluids* 26 (2) (2014).
- [17] S. Le Clainche, J.M. Vega, Higher order dynamic mode decomposition, *SIAM J. Appl. Dyn. Syst.* 16 (2) (2017) 882–925.
- [18] P.J. Baddoo, B. Herrmann, B.J. McKeon, J. Nathan Kutz, S.L. Brunton, Physics-informed dynamic mode decomposition, *Proc. R. Soc. A* 479 (2271) (2023) 20220576.
- [19] P.J. Schmid, Dynamic mode decomposition and its variants, *Annu. Rev. Fluid Mech.* 54 (1) (2022) 225–254.
- [20] P. Benner, S. Grivet-Talocia, A. Quarteroni, G. Rozza, W. Schilders, L.M. Silveira, *System-and Data-Driven Methods and Algorithms*, De Gruyter, 2021.
- [21] Q.A. Huhn, M.E. Tano, J.C. Ragusa, Y. Choi, Parametric dynamic mode decomposition for reduced order modeling, *J. Comput. Phys.* 475 (2023) 111852.
- [22] T. Sayadi, P.J. Schmid, F. Richecoeur, D. Durox, Parametrized data-driven decomposition for bifurcation analysis, with application to thermo-acoustically unstable systems, *Phys. Fluids* 27 (3) (2015).
- [23] D. Amsallem, *Interpolation on Manifolds of CFD-Based Fluid and Finite Element-Based Structural Reduced-Order Models for On-Line Aeroelastic Predictions*, Stanford University, 2010.
- [24] M.W. Hess, A. Quaini, G. Rozza, A data-driven surrogate modeling approach for time-dependent incompressible Navier-Stokes equations with dynamic mode decomposition and manifold interpolation, *Adv. Comput. Math.* 49 (2) (2023) 22.
- [25] K. Li, Y. Li, L. Li, S. Lanteri, Surrogate modeling of time-domain electromagnetic wave propagation via dynamic mode decomposition and radial basis function, *J. Comput. Phys.* 491 (2023) 112354.
- [26] Z. Gao, Y. Lin, X. Sun, X. Zeng, A reduced order method for nonlinear parameterized partial differential equations using dynamic mode decomposition coupled with k-nearest-neighbors regression, *J. Comput. Phys.* 452 (2022) 110907.
- [27] H. Song, Y. Ba, D. Chen, Q. Li, A model reduction method for parametric dynamical systems defined on complex geometries, *J. Comput. Phys.* 506 (2024) 112923.
- [28] G.H. Golub, C.F. Van Loan, *Matrix Computations*, JHU press, 2013.
- [29] T. Lieu, M. Lesoinne, Parameter adaptation of reduced order models for three-dimensional flutter analysis, in: *42nd AIAA Aerospace Sciences Meeting and Exhibit*, 2004, p. 888.
- [30] T. Lieu, C. Farhat, Aerodynamic parameter adaptation of CFD-based reduced-order models, in: *45th AIAA Aerospace Sciences Meeting and Exhibit*, 2007, p. 328.
- [31] D. Amsallem, C. Farhat, Interpolation method for adapting reduced-order models and application to aeroelasticity, *AIAA J.* 46 (7) (2008) 1803–1813.
- [32] T. Krake, D. Weiskopf, B. Eberhardt, Dynamic mode decomposition: theory and data reconstruction, [arXiv:1909.10466](https://arxiv.org/abs/1909.10466) (2019).
- [33] P.J. Schmid, Application of the dynamic mode decomposition to experimental data, *Exp. Fluids* 50 (2011) 1123–1130.
- [34] J.H. Tu, *Dynamic Mode Decomposition: Theory and Applications*, Ph.D. thesis, Princeton University, 2013.
- [35] T. Sayadi, P.J. Schmid, J.W. Nichols, P. Moin, Reduced-order representation of near-wall structures in the late transitional boundary layer, *J. Fluid Mech.* 748 (2014) 278–301.
- [36] H. Lu, D.M. Tartakovsky, Model reduction via dynamic mode decomposition, [arXiv:2204.09590](https://arxiv.org/abs/2204.09590) (2022).

- [37] P.-A. Absil, R. Mahony, R. Sepulchre, Riemannian geometry of Grassmann manifolds with a view on algorithmic computation, *Acta Appl. Math.* 80 (2004) 199–220.
- [38] Z.-H. Wan, L. Zhou, B.-F. Wang, D.-J. Sun, Dynamic mode decomposition of forced spatially developed transitional jets, *Eur. J. Mech. B/Fluids* 51 (2015) 16–26.
- [39] P.J. Schmid, D. Violato, F. Scarano, Decomposition of time-resolved tomographic PIV, *Exp. Fluids* 52 (2012) 1567–1579.
- [40] G. Tissot, L. Cordier, N. Benard, B.R. Noack, Model reduction using dynamic mode decomposition, *Comptes Rendus. Mécanique* 342 (6–7) (2014) 410–416.
- [41] J. Kou, W. Zhang, An improved criterion to select dominant modes from dynamic mode decomposition, *Eur. J. Mech. B/Fluids* 62 (2017) 109–129.
- [42] L. Xu, Z. Liu, X. Li, M. Zhao, Y. Zhao, An improved mode time coefficient for dynamic mode decomposition, *Phys. Fluids* 35 (10) (2023).
- [43] A. Weiner, R. Semaan, Robust dynamic mode decomposition methodology for an airfoil undergoing transonic shock buffet, *AIAA J.* 61 (10) (2023) 4456–4467.
- [44] A. Edelman, T.A. Arias, S.T. Smith, The geometry of algorithms with orthogonality constraints, *SIAM J. Matrix Anal. Appl.* 20 (2) (1998) 303–353.
- [45] D.G. Giovanis, M.D. Shields, High-dimensional interpolation on the Grassmann manifold using Gaussian processes, in: 13th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP13), 2019, pp. 1041–1047.
- [46] A. Galántai, C.J. Hegedűs, Jordan’s principal angles in complex vector spaces, *Numer. Linear Algebr. Appl.* 13 (7) (2006) 589–598.
- [47] T. Bendokat, R. Zimmermann, P.-A. Absil, A Grassmann manifold handbook: basic geometry and computational aspects, *Adv. Comput. Math.* 50 (1) (2024) 6.
- [48] L.M. Yang, X. Zhao, C. Shu, Y.J. Du, Parametric reduced order modeling-based discrete velocity method for simulation of steady rarefied flows, *J. Comput. Phys.* 430 (2021) 110037.
- [49] E. Kasner, Conformality in connection with functions of two complex variables, *Trans. Am. Math. Soc.* 48 (1) (1940) 50–62.
- [50] E. Kasner, J. DeCicco, Pseudo-conformal geometry: functions of two complex variables, *Bull. Amer. Math. Soc.* 48, (1942) 317–328.
- [51] K. Scharnhorst, Angles in complex vector spaces, *Acta Appl. Math.* 69 (2001) 95–103.
- [52] M. Oulghelou, C. Allery, Non intrusive method for parametric model order reduction using a bi-calibrated interpolation on the Grassmann manifold, *J. Comput. Phys.* 426 (2021) 109924.
- [53] Y. Delaunay, Control of Circular Cylinder Wakes Using Base Mass Actuation, Ph.D. thesis, ETH Zurich, 2000.
- [54] F. Caruso Lombardi, A. Bongarzone, G.A. Zampogna, F. Gallaire, S. Camarri, P.G. Ledda, Von Kármán vortex street past a permeable circular cylinder: two-dimensional flow and dynamic-mode-decomposition-based secondary stability analysis, *Phys. Rev. Fluids* 8 (8) (2023) 083901.
- [55] D. Massaro, P. Schlatter, Global stability of the flow past a stepped cylinder, *J. Fluid Mech.* 988 (2024) A1.
- [56] R. Bourguet, Flow-induced vibrations of a cylinder along a circular arc, *J. Fluid Mech.* 954 (2023) A7.
- [57] M.O. Deville, P.F. Fischer, E.H. Mund, *High-Order Methods for Incompressible Fluid Flow*, 9, Cambridge university press, 2002.
- [58] A.T. Patera, A spectral element method for fluid dynamics: laminar flow in a channel expansion, *J. Comput. Phys.* 54 (3) (1984) 468–488.
- [59] B. Fornberg, *A Practical Guide to Pseudospectral Methods*, 1, Cambridge university press, 1998.
- [60] G.E. Karniadakis, S.J. Sherwin, *Spectral/hp Element Methods for CFD*, Oxford University Press, 1999.
- [61] J.L. Callahan, J.-C. Loiseau, S.L. Brunton, Multiscale model reduction for incompressible flows, *J. Fluid Mech.* 973 (2023) A3.
- [62] G. Ciaramella, M.J. Gander, T. Vanzan, A gentle introduction to interpolation on the Grassmann manifold, Oberwolfach Preprint, Mathematisches Forschungsinstitut Oberwolfach, 2024, <https://doi.org/10.14760/OWP-2024-02>.
- [63] S.-s. Chern, *Complex Manifolds without Potential Theory: with an appendix on the geometry of characteristic classes*, Springer, 1979.
- [64] J.H. Manton, Optimization algorithms exploiting unitary constraints, *IEEE Trans. Signal Process.* 50 (3) (2002) 635–650.
- [65] E. Estrada, Geometries of mixed graphs in complex vector spaces. Hierarchies and clusters in complex networks, *Math. Model. Nat. Phenom.* 20 (2025) 7.